

Work Breakdown Structure (WBS) Specification

Domain Allocation & Work Packages: Interface Software Subsystem

Framework: VDI 2206 Mechatronic System Design (Descending Branch Phase 2 / ISO 13482 Safety Guidelines)

Mechatronics Engineering Graduation Project | Technical Report WP-SW-04 | October 2026

Abstract

This engineering specification defines the formal Work Breakdown Structure (WBS) for the **Interface Software Subsystem** (WP-SW-04) of the modular semi-humanoid robotic platform. In compliance with the **VDI 2206** design methodology for mechatronic and cyber-physical systems (CPS), the software domain is decomposed into six canonical pillars: Navigation (Nav2), Arm Control (MoveIt 2), Machine Vision, Interface, Vision-Language-Action (VLA), and Low-Level Firmware (`micro_ros` + `ros2_control`). Within this hierarchy, the Interface Subsystem bridges high-level cognitive models, deterministic control middleware, human operators, and distributed edge networks. This document details the Level-2 through Level-4 decomposition of the five interface pillars: (1) **Browser Teleoperation & Web Dashboard**, (2) **Standardized Task Port Gateway**, (3) **Autonomous Skill Marketplace & Registry**, (4) **Physical & Ambient HMI**, and (5) **Developer Client SDK & API Gateways**. For each work package, technical scopes, engineering deliverables, interface contracts, and quantifiable acceptance verification metrics are rigorously formulated.

1 System Context & VDI 2206 Architectural Integration

In accordance with the **VDI 2206** V-model for multidisciplinary mechatronic engineering, the system design phase maps functional requirements into decoupled domain-specific components: Mechanical, Electrical/Embedded, and Computer Science/Software. The software domain is structured as a multi-rate, contract-driven architecture that rigorously decouples probabilistic cognition from hard real-time physical control:

$$\mathcal{S}_{\text{Software}} = \left\{ \mathcal{D}_{\text{NAV2}}, \mathcal{D}_{\text{MoveIt2}}, \mathcal{D}_{\text{Vision}}, \mathcal{D}_{\text{Interface}}, \mathcal{D}_{\text{VLA}}, \mathcal{D}_{\text{LowLevel}} \right\}. \quad (1)$$

Within this topology, the **Interface Subsystem** ($\mathcal{D}_{\text{Interface}}$, designated as **WBS 4.0**) serves as the mandatory mediator between external agents (human operators, fleet management servers, third-party developers, on-chassis touchpoints) and internal robotic engines. To guarantee absolute compliance with the **Command Contract** and ISO 13482 personal care robotics safety guidelines, no external interface may directly write to raw actuator registers, motor current loops, or low-level PWM buses. Instead, all interaction flows across validated action servers, projection filters, and sandboxed execution runtimes.

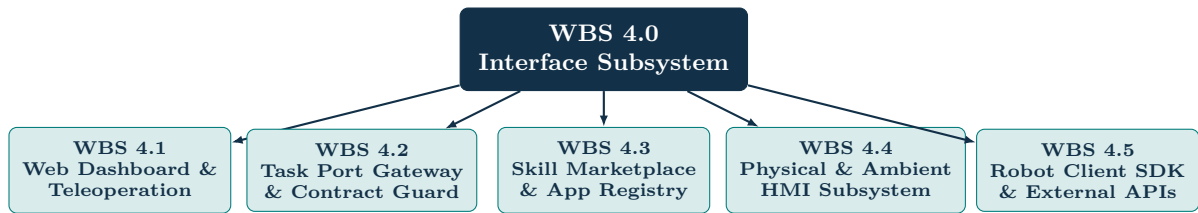


Figure 1: Level-2 structural decomposition of the Interface Software Subsystem ($\mathcal{D}_{\text{Interface}}$) under the VDI 2206 framework.

2 Hierarchical Work Breakdown Structure (Level 2 to Level 4)

The Interface Subsystem is decomposed into 5 Level-2 work elements, 17 Level-3 work packages, and 45 Level-4 discrete engineering work tasks.

2.1 WBS 4.1: Browser Teleoperation & Web Fleet Dashboard

Objective: Provide a zero-install, responsive single-page web console (React / TypeScript / Vite / TailwindCSS) enabling remote situational awareness, glass-to-glass low-latency video streaming, bilateral joystick/gamepad teleoperation, interactive point cloud visualization, and flight telemetry logging.

- **WBS 4.1.1: WebRTC Real-Time Video Streaming Subsystem**

- *4.1.1.1 H.264 GStreamer Hardware Acceleration Pipeline:* Configure Jetson Orin NVENC hardware encoder plugins to capture RealSense stereo RGB-D streams (1280×720 @ 30 fps) with glass-to-glass latency $t_{g2g} \leq 80$ ms.
- *4.1.1.2 WebRTC Signaling & NAT Traversal Gateway:* Implement WebSocket signaling daemon with STUN/TURN traversal fallback for secure WAN/LAN teleoperation.
- *4.1.1.3 Multi-Camera HUD Compositor:* Ingest auxiliary wrist and neck RGB streams into selectable dashboard picture-in-picture (PiP) viewport layouts.

- **WBS 4.1.2: Dual-Protocol Middleware Bridge (WebSocket / Zenoh-Bridge-Web)**

- *4.1.2.1 Rosbridge Suite Integration & QoS Rate Decimator:* Bridge ROS 2 topics, services, and actions to browser WebSockets with dynamic topic rate throttling (downsampling 100 Hz state topics to 20 Hz browser updates to conserve network bandwidth).
- *4.1.2.2 Zenoh-Bridge-Web Ultra-Low Latency Transport:* Provide native WebAssembly (Wasm) Zenoh client bindings for peer-to-peer telemetry streaming bypassing JSON serialization overhead.
- *4.1.2.3 Token-Based Authentication & RBAC Interceptor:* Enforce JWT/mTLS token authorization for teleoperator session establishment and operator privilege escalation.

- **WBS 4.1.3: Teleoperation & Input Controller Layer**

- *4.1.3.1 HTML5 Gamepad API Driver:* Support dual-analog industrial/consumer gamepads (e.g., Logitech F710, Xbox) with deadband calibration, exponential throttle curves, and button mapping profiles.
- *4.1.3.2 Virtual Touch Joysticks & Mobile Responsive UI:* Render touch-sensitive on-screen virtual joysticks for tablet/smartphone operation with emergency release deadman switch.
- *4.1.3.3 End-Effector Cartesian Jogging Console:* Interactive 6-DoF Cartesian jog widgets for incremental millimeter-precision tool-flange positioning and tool orientation adjustments.

- **WBS 4.1.4: 3D Digital Twin & Sensor Data Visualization**

- *4.1.4.1 Three.js / WebGL URDF Robot Model Renderer:* Real-time 3D synchronized rendering of the 18-DoF robot kinematic chain driven by live `/joint_states` topic transforms.
- *4.1.4.2 2D LiDAR Costmap & Nav2 Goal Pose Interactor:* Stream 2D occupancy grids and local costmaps with click-and-drag interactive navigation goal and orientation placement tools.
- *4.1.4.3 Point Cloud Downsampler & Octree Visualizer:* WebGL point cloud shader visualizing RealSense 3D point clusters with client-side voxel downsampling ($< 150k$ points/frame).

- **WBS 4.1.5: System Health, Telemetry & Flight Recorder Dashboard**

- *4.1.5.1 Power Domain & Battery BMS Gauge Console:* Real-time visual display of cell voltages, State of Charge (SoC%), power draw (W), and thermal status across the 24V, 19V, 12V, and 5V rails.
- *4.1.5.2 Motor Actuator Diagnostic Matrix:* Grid monitor for all $9 \times$ CubeMars AK70-10 QDDs and $6 \times$ Feetech servos showing winding temperature, phase current, bus voltage, and error flags.
- *4.1.5.3 ROS 2 Node Lifecycle Inspector & Diagnostic Logger:* Interactive console visualizing ROS 2 Lifecycle Node transitions, active fault logs, and historical diagnostic bag downloads.

2.2 WBS 4.2: Standardized Task Port Gateway & Contract Guard

Objective: Provide a strictly typed, versioned ROS 2 Action and Service gateway that isolates low-level real-time controllers from external cognitive models and users, enforcing the mathematical Command Contract and ISO 13482 safety envelope.

- **WBS 4.2.1: High-Level Primitive Action Server Implementations**

- *4.2.1.1 Navigation Task Port (`/navigate_to_pose`):* Wrap Nav2 navigation client into an observable action interface supporting goal preemption, obstacle wait timeouts, and dynamic recovery behaviors.
- *4.2.1.2 Tabletop Manipulation Task Port (`/pick_object`, `/place_object`):* Expose semantic object pick/place primitives accepting detected object bounding boxes, automatically generating grasp poses, pre-approach trajectories, and compliant grip triggers.

- 4.2.1.3 *Autonomous Docking Task Port (/dock_robot)*: Closed-loop docking primitive aligning with charging contacts via infrared and LiDAR retro-reflector feedback.
- 4.2.1.4 *Bimanual Coordinated Carry Task Port (/carry_dual_arm)*: Synchronous dual-arm Cartesian action enforcing closed kinematic chain constraints during payload transport.
- **WBS 4.2.2: Mathematical Command Contract Verification & Projection Filter**
 - 4.2.2.1 *Feasible Set Projector ($\Pi_{\mathcal{F}}$ Operator)*: Quadratic programming solver projecting incoming external trajectory deltas into the verified feasible space $\mathcal{F}(\mathbf{q}, \mathcal{E}, \mathcal{S})$, clamping velocities, accelerations, and workspace boundaries.
 - 4.2.2.2 *Observable Safety Projection Veto Engine*: Continuous residual monitoring ($\varepsilon_{\text{proj}} > \varepsilon_{\text{tol}}$). Automatically veto unsafe AI proposals, emit structured **SafetyVeto** warnings, and trigger smooth kinematic deceleration stops within ≤ 20 ms.
 - 4.2.2.3 *ISO 13482 Dynamic Speed & Separation Monitor (SSM)*: Dynamically scale max tool and base velocity inversely with human proximity detected via 2D LiDAR and depth camera fields.
- **WBS 4.2.3: Task State Machine & Execution Supervisor**
 - 4.2.3.1 *BehaviorTree.CPP Execution Root Bridge*: Translate incoming Task Port action goals into parameterized Behavior Tree blackboard variables and supervise tree ticking lifecycle.
 - 4.2.3.2 *Preemption, Cancellation & Pause Handler*: Deterministic cancellation handler safely transitioning manipulators and base to locked stationary states upon operator override or external preemption.
 - 4.2.3.3 *Structured Failure Diagnostic Code Generator*: Output standardized error classification schemas (e.g., `ERRИК_UNREACHABLE`, `ERR_PATH_BLOCKED`, `ERR_GRASP_SLIP`) for automated upstream replanning.

2.3 WBS 4.3: Autonomous Skill Marketplace & Application Registry

Objective: Provide a modular, secure runtime repository and lifecycle orchestrator for packaging, distributing, dynamically loading, and executing modular robot skills, AI policies, and composite behavior trees without recompiling system software.

- **WBS 4.3.1: Standardized Robot Skill Packaging Format**
 - 4.3.1.1 *Skill Package Schema Definition (OpenUSD / JSON-Schema)*: Manifest specification declaring skill entry points, input/output parameters, required hardware interfaces, pre/post-conditions, and execution safety envelopes.
 - 4.3.1.2 *Deterministic Behavior Tree XML Bundler*: Containerized packaging of BehaviorTree.CPP XML subtrees, node plugins, and associated ROS 2 parameter YAML configurations.
 - 4.3.1.3 *Policy Checkpoint & Neural Weight Artifacts*: Standardized storage format for ONNX/TensorRT policy weights (e.g., OpenVLA affordance chunks, tactile slip models) with SHA-256 integrity verification.
- **WBS 4.3.2: On-Robot Skill Engine & Dynamic Plugin Loader**
 - 4.3.2.1 *Shared Library PluginLoader Service*: Dynamic runtime loading and unloading of compiled C++ BehaviorTree action nodes and MoveIt planning plugins without restarting core ROS 2 daemons.
 - 4.3.2.2 *Skill Dependency Validator & Capability Checker*: Pre-flight introspection verifying that required sensors, joint limits, end-effectors, and transform frames are online and calibrated before initiating execution.
 - 4.3.2.3 *Sandboxed Execution Environment*: Micro-container / cgroups process jail isolating experimental third-party skills, enforcing CPU, memory, and bus bandwidth quotas.
- **WBS 4.3.3: Marketplace Cloud / Local Registry & Package Manager**
 - 4.3.3.1 *Local Skill Repository Web UI*: Web Dashboard catalog view allowing operators to search, browse, install, inspect parameters, and execute installed platform skills.
 - 4.3.3.2 *Cloud Sync & Over-The-Air (OTA) Release Client*: Secure HTTPS/gRPC client synchronizing approved enterprise skill bundles, verified firmware revisions, and cryptographic license tokens.
 - 4.3.3.3 *Graphical Behavior Tree Macro Composer*: Web-based visual drag-and-drop workflow editor enabling users to chain discrete skills into custom sequential and parallel mission trees.

2.4 WBS 4.4: Physical, Acoustic & Ambient HMI Subsystem

Objective: Implement the multimodal physical human-robot interface on the robot body, including the torso touchscreen GUI, ambient expression LED indicator, directional speech synthesis/recognition, and hardware service bulkhead.

- **WBS 4.4.1: Torso Embedded Touchscreen GUI (Kiosk Mode)**

- *4.4.1.1 Lightweight On-Chassis UI (Flutter / Qt Embedded):* 5–7” touch display application running in kiosk mode with high-contrast outdoor/indoor UI themes and glove-friendly touch targets.
- *4.4.1.2 Quick-Diagnostic & Network Configuration View:* Immediate display of IP addresses, Wi-Fi connectivity wizard, battery SoC%, hardware temperature readouts, and E-Stop latch status.
- *4.4.1.3 On-Site Pin-Protected Task Dispatcher:* Local interface enabling facility workers to select and dispatch predefined local routines (e.g., “Go to Cleaning Bay”, “Handover Parcel”) without requiring a laptop.

- **WBS 4.4.2: Expressive Ambient Lighting & Head Audio HMI**

- *4.4.2.1 Addressable RGB LED Ring Controller:* ESP32-driven WS2812B NeoPixel ring mounted on the head perimeter displaying unambiguous system states (e.g., solid green: idle ready; pulsing blue: autonomous motion; blinking amber: obstacle pause; flashing red: E-Stop).
- *4.4.2.2 Directional Audio & Head Gaze Tracker:* Coordinate ReSpeaker dual-mic array TDOA sound localization with the 2-DoF pan/tilt neck servos, autonomously turning the robot head toward an interacting human speaker.
- *4.4.2.3 Text-To-Speech (TTS) & Acoustic Feedback Daemon:* Onboard Piper/VITS neural speech synthesizer generating low-latency (< 200 ms) natural voice confirmations and audible safety alarm chimes over the 2W internal speaker.

- **WBS 4.4.3: Physical Bulkhead Service Bay & Mechanical HMI**

- *4.4.3.1 Hardware Interlock & E-Stop UI Monitoring:* Sense NC loop contacts of the torso mushroom E-Stop button and report trip timestamp and circuit resistance via digital GPIO optocouplers.
- *4.4.3.2 Bulkhead USB/RJ45 Maintenance Daemon:* Automatically detect maintenance cable plug-in on the torso service bulkhead, spawning isolated tethered development networks and auto-mounting diagnostic flash media.

2.5 WBS 4.5: Developer Client SDK & External Integration Gateways

Objective: Provide native client libraries, RESTful/WebSocket APIs, and industrial fleet adapters (VDA 5050 / MQTT) to enable seamless third-party software integration, academic research scripting, and warehouse management integration.

- **WBS 4.5.1: High-Level Python Client SDK (`robot_sdk`)**

- *4.5.1.1 Ergonomic Python Asynchronous API:* Provide intuitive Pythonic classes (Robot, Arm, Base, Gripper, Head) wrapping complex ROS 2 actions into `asyncio` Python promises.
- *4.5.1.2 Kinematics & Transform Client Utilities:* Local client-side forward kinematics helpers, coordinate frame transforms ($SE(3)$ representations), and spatial vector math utilities.
- *4.5.1.3 Demonstration Recording & Replay Toolkit:* CLI tool to record synchronous streams of joint positions, gripper states, and RGB-D images into HDF5/Zarr formats for imitation learning and dataset curation.

- **WBS 4.5.2: RESTful & WebSocket API Gateway Server**

- *4.5.2.1 OpenAPI / Swagger REST Endpoints (FastAPI):* Asynchronous HTTP gateway exposing system status, telemetry queries, skill listings, configuration updates, and mission dispatching for web developers.
- *4.5.2.2 Push-Event WebSocket Telemetry Streamer:* High-throughput JSON/MessagePack event stream pushing asynchronous state changes, battery events, and task completion notices to subscribed client applications.
- *4.5.2.3 Security Gateway (OAuth2 / TLS 1.3 Encryption):* Strict end-to-end cryptographic encryption with API key generation, rate limiting, and automated certificate renewal.

- **WBS 4.5.3: Industrial Enterprise & Logistics Adapters**

- 4.5.3.1 *VDA 5050 Fleet Interface Adapter*: Native translation bridge converting AGV/AMR standard VDA 5050 order topics and instant actions into Nav2 goal dispatches and state telemetry reports.
- 4.5.3.2 *Industrial MQTT IoT Bridge*: Lightweight telemetry publishing client transmitting machine operating hours, energy consumption, and fault indicators to factory IoT brokers.

3 Work Package Specification & Allocation Matrix

Table 1 formalizes the engineering allocation matrix for the Interface Subsystem in strict compliance with the project’s VDI 2206 requirements, identifying responsible engineering sub-disciplines, explicit work package deliverables, cross-domain interface boundaries, and quantifiable verification criteria.

Table 1: VDI 2206 engineering work package allocation matrix for Interface Software Subsystem (WP-SW-04).

WBS Code	Work Package Title	Scope & Responsibilities	Key	Primary Deliverables	Verification Metric
4.1.1	WebRTC Video Teleop Pipeline	GStreamer NVENC hardware encoding of RealSense color/depth streams, WebRTC signaling server, NAT traversal.		C++ ROS 2 WebRTC streamer node, signaling daemon, browser video client component.	Glass-to-glass latency $t_{g2g} \leq 80$ ms; frame rate ≥ 30 fps at 720p; CPU load $< 10\%$.
4.1.2	WebSocket / Zenoh Bridge	Two-way state telemetry transport, dynamic topic decimation, JWT token authentication, client session handling.		Rosbridge configuration packages, Zenoh-Web Wasm module, auth middleware plugin.	Telemetry rate ≥ 20 Hz; round-trip command latency ≤ 25 ms; zero unauthorized packet leaks.
4.1.3	Gamepad & Jogging Layer	HTML5 gamepad input mapping, deadband compensation, mobile touch joysticks, 6-DoF Cartesian jog widgets.		React controller hooks, virtual joystick UI component, Cartesian jog action client.	Input dispatch jitter < 5 ms; deadman disconnect stop latency ≤ 50 ms.
4.1.4	3D Digital Twin Visualizer	Three.js URDF model rendering, live TF frame synchronization, 2D Nav2 costmap overlay, point cloud shaders.		WebGL canvas component, TF tree subscriber, point cloud downsampling worker.	Render frame rate ≥ 60 fps; point cloud capacity $\geq 150k$ points; transform error 0 mm.
4.1.5	Diagnostics & Flight Logger	Power rail BMS telemetry display, motor actuator thermal/current gauges, ROS 2 node lifecycle dashboard.		Diagnostic dashboard UI, real-time alert toast notification system, bag exporter.	Update latency ≤ 100 ms; 100% fault coverage across all 15 active actuators.
4.2.1	Primitive Task Action Servers	High-level action interfaces for navigation, tabletop pick/place, autonomous docking, and bimanual carry.		ROS 2 Action packages (<code>NavigateToPose</code> , <code>PickObject</code> , <code>Dock</code>), client test suites.	Action dispatch latency ≤ 25 ms; preemption reaction time ≤ 15 ms; 100% parameter typing.

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WBS Code	Work Package Title	Scope & Responsibilities	Key	Primary Deliverables	Verification Metric
4.2.2	Command Contract Guard	Π -projection quadratic operator, velocity/acceleration safety veto handler, ISO 13482 SSM.		C++ real-time lifecycle guard node, projection optimization library, unit test harness.	Projection solve time ≤ 2 ms; safety veto trigger time ≤ 20 ms; zero unprojected writes.
4.2.3	Task Execution Supervisor	BehaviorTree.CPP engine wrapper, blackboard variable injection, preemption/pause handlers, error code schemas.		BT execution engine package, lifecycle supervisor node, structured error definition headers.	State transition deterministic latency < 5 ms; zero deadlocks on sub-tree aborts.
4.3.1	Skill Packaging Format	Specification of skill schema, BehaviorTree XML packaging, ONNX/TensorRT model weight artifactory.		Skill manifest JSON schema, packaging CLI tool (skill-pack), verification test suites.	Schema validation time < 50 ms; cryptographic hash verification 100% tamper-proof.
4.3.2	Dynamic Plugin Loader	Runtime C++ plugin loading via <code>pluginlib</code> , pre-flight hardware dependency checker, sandboxed cgroups jail.		Dynamic skill runner node, pre-flight inspection script, cgroups isolation configuration.	Plugin load time < 500 ms; zero core daemon crashes upon bad plugin abort.
4.3.3	Marketplace & Composer	Local skill catalog UI, cloud sync client, visual drag-and-drop Behavior Tree composer.		Marketplace web application, OTA sync client daemon, graphical web node editor.	UI tree compilation to valid XML < 100 ms; OTA rollback capability ≤ 10 s.
4.4.1	Torso Touchscreen Kiosk	5–7" touch display UI in kiosk mode, network config wizard, battery/error monitor, local task trigger.		Touchscreen UI application (Flutter/Qt), systemd kiosk service, pin authentication module.	Cold boot to touch response ≤ 8 s; UI frame rate ≥ 60 fps; touch latency < 30 ms.
4.4.2	Expressive Lighting & Audio	WS2812B NeoPixel ring state animations, ReSpeaker TDOA head gaze tracking, Piper neural TTS voice synthesizer.		ESP32 LED firmware, head gaze tracking node, local TTS audio generation service.	Sound localization angle accuracy $\pm 10^\circ$; gaze tracking response < 300 ms; TTS latency < 200 ms.
4.4.3	Physical Bulkhead Bay	Hardware E-Stop status sensing via optocouplers, bulkhead USB/GbE automatic network tether daemon.		Optocoupler sensing driver, auto-tether udev scripts, maintenance terminal launcher.	E-Stop state detection latency ≤ 5 ms; tether network acquisition time < 3.0 s.
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WBS Code	Work Package Title	Scope & Responsibilities	Key	Primary Deliverables	Verification Metric
4.5.1	Python Client SDK	Async Python library wrapping Task Port actions, kinematics utilities, HDF5 demonstration recording tools.		<code>robot_sdk</code> package (wheel), Sphinx documentation, developer code examples.	API documentation test coverage $\geq 90\%$; action dispatch round-trip < 10 ms.
4.5.2	REST & WebSocket Gateway	FastAPI REST endpoints for fleet queries, push-event WebSocket telemetry server, OAuth2 authentication.		FastAPI gateway service container, OpenAPI Swagger UI, JWT security layer.	REST endpoint response time $t_{99} < 15$ ms; concurrent WebSocket connections ≥ 50 .
4.5.3	Industrial Logistics Adapters	VDA 5050 AGV/AMR protocol translator, industrial MQTT broker bridge for factory IoT telemetry.		VDA 5050 ROS 2 node, MQTT publisher gateway, integration test harness.	VDA 5050 compliance certified; message translation latency < 10 ms; zero message drops.

4 Cross-Subsystem Interface Boundaries & Data Contracts

To avoid integration failure modes during Phase 3 of the VDI 2206 model (System Integration), all boundaries between the Interface Subsystem and adjacent subsystems are governed by immutable data contracts:

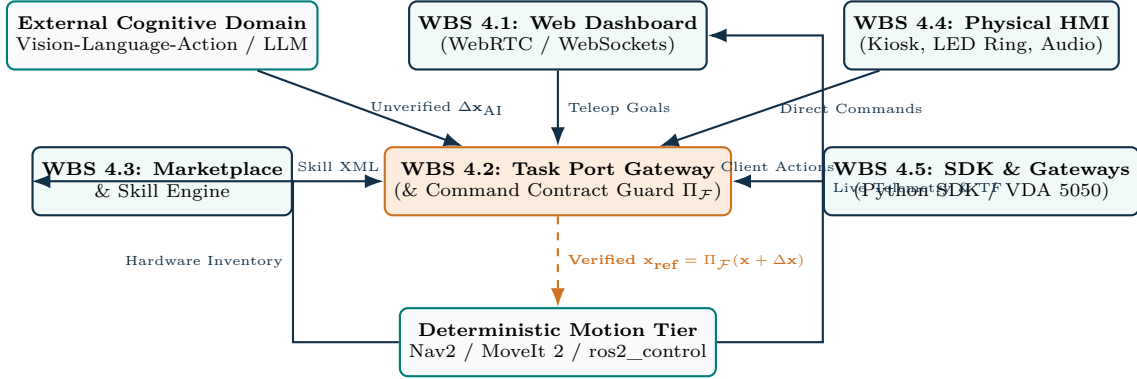


Figure 2: Cross-subsystem data communication and contract enforcement topology.

- Interface 4-A (Interface $\rightarrow$ Motion Planning [Nav2 / MoveIt 2]):** All spatial trajectory and waypoint goals are committed exclusively via ROS 2 action protocols (`nav2_msgs/action/NavigateToPose`, `moveit_msgs/action/MoveGroup`). The Command Contract Guard guarantees that target goal poses reside within collision-free configuration space prior to planner invocation.
- Interface 4-B (Interface $\rightarrow$ Machine Vision & Perception):** High-bandwidth sensory streams (uncompressed 1280×720 RGB-D frames, segmented point clouds) are consumed via zero-copy shared memory transports (`sensor_msgs/Image`, `sensor_msgs/PointCloud2`). The WebRTC pipeline taps into hardware NVENC encoders directly without burdening the host CPU.
- Interface 4-C (Interface $\rightarrow$ Low-Level Firmware [micro_ros + ros2_control]):** Bidirectional exchange of raw actuator states and commanded joint efforts. Interface 4.1 reads diagnostic registers (winding temperature, current draw, encoder health) across `diagnostic_msgs/DiagnosticArray` topics at 10 Hz, completely isolated from the 1 kHz real-time fieldbus loop.

4. **Interface 4-D (Interface $\rightarrow$ VLA Foundation Policy Domain):** The VLA inference policy dispatches proposed spatial deltas $\Delta \mathbf{x}_{AI}$ via the `/vla/propose_action` service. The Task Port Gateway intercepts this unverified vector, subjects it to the quadratic projection $\Pi_{\mathcal{F}}$, and passes only bounded setpoints downstream.

5 Verification, Testing & Acceptance Roadmap

In adherence to the ascending branch of the VDI 2206 V-model, verification proceeds through three sequential testing gates prior to operational deployment:

- **Gate 1: Software-in-the-Loop (SIL) Virtual Validation:** The entire Interface Subsystem is deployed inside containerized Docker environments interfaced to an NVIDIA Isaac Sim / Gazebo physics simulation. Automated CI/CD test suites inject synthetic network degradation (packet dropouts up to 15%, latency jitter up to 150 ms) to verify WebRTC reconnection robustness, action cancellation determinism, and zero UI crash occurrences.
- **Gate 2: Hardware-in-the-Loop (HIL) Testbench Validation:** The interface components are deployed onto the physical NVIDIA Jetson Orin NX compute module paired with the physical Torso Touchscreen and ReSpeaker audio head. Bench testing verifies that glass-to-glass video latency remains strictly ≤ 80 ms, CPU utilization remains below 15%, and sound localization accuracy achieves $\pm 10^\circ$.
- **Gate 3: Full Platform Integrated Safety & ISO 13482 Validation:** On the integrated mobile manipulator, physical hazard injection tests are performed: triggering manual E-Stop while teleoperating, simulating sudden obstacle occlusion during waypoint jogging, and injecting out-of-bounds trajectory proposals from the VLA model. Acceptance requires that the safety projection veto clamps the command within ≤ 20 ms and brings the platform to a controlled stop without mechanical stress.

6 Conclusion

The Work Breakdown Structure developed herein establishes an exhaustive, mathematically grounded, and modular engineering specification for the Interface Software Subsystem ($\mathcal{D}_{\text{Interface}}$). By decomposing the interface into five decoupled pillars—Web Dashboard, Task Port Gateway, Skill Marketplace, Physical HMI, and Developer SDK—the platform satisfies the functional requirements of modern semi-humanoid robotics: agile teleoperation, secure cognitive AI integration, intuitive human collaboration, and enterprise scalability, while maintaining an uncompromising commitment to ISO 13482 physical safety.